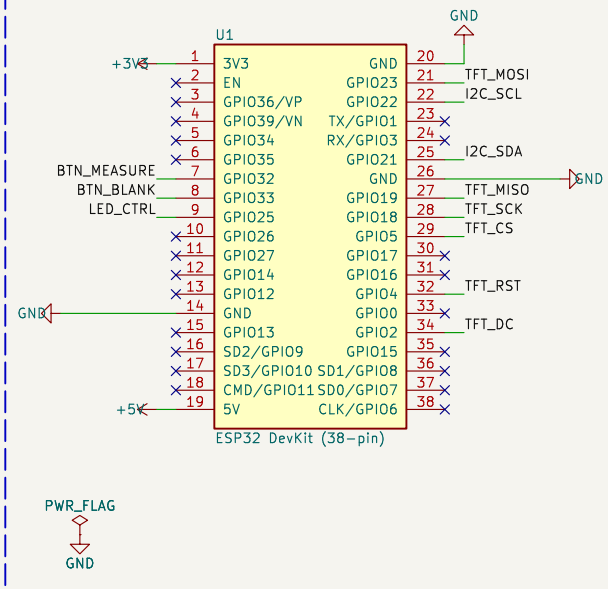
1. ESP32 DEVKIT (the brain)

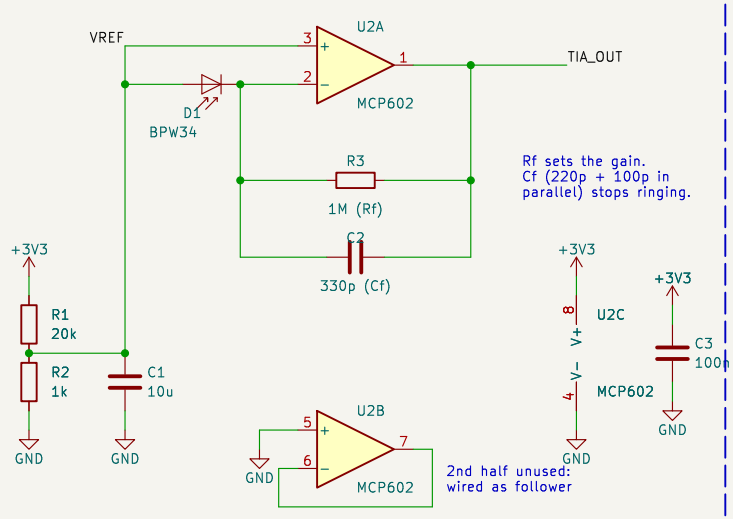
Powered from USB. Gives 3.3 V and 5 V to the rest.



2. LIGHT -> VOLTAGE (photodiode + transimpedance amplifier)

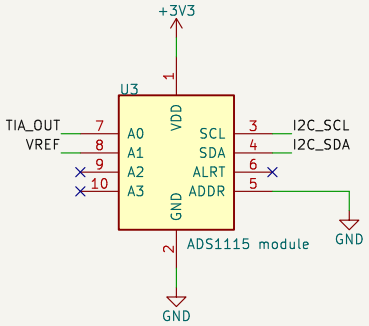
Vout = Vref + I(photo) x Rf 1 nA of light current -> 1 mV

Vref = 3.3 V x 1k / 21k = 0.157 V
keeps the op-amp output away from 0 V



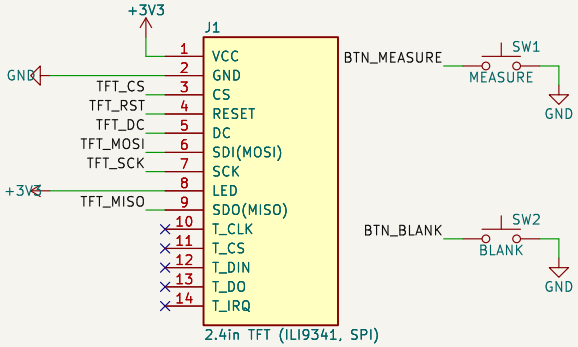
3. VOLTAGE -> NUMBER

ADS1115 16-bit ADC module, I2C 0x48



Reads A0 - A1 (differential):
the TIA output measured
against Vref, so the 0.157 V
cancels out.

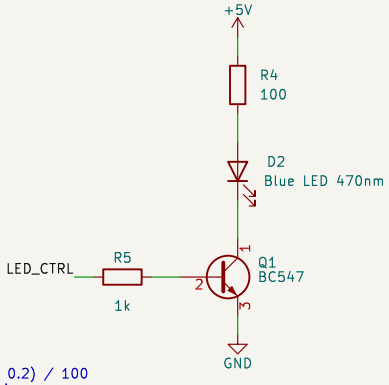
5. DISPLAY + BUTTONS



Buttons use the ESP32
internal pull-ups

4. EXCITATION LED

GPIO25 HIGH -> transistor on -> LED on



I(LED) = (5 - 3.0 - 0.2) / 100
= about 18 mA

BUILD NOTES
- Same-name labels are connected (e.g. VREF, TIA_OUT, I2C_SDA).
- Keep D1, R3, C2 and U2 pins 1-3 close together, short wires.
- D1 looks at the cuvette at 90 deg to the LED, behind a long-pass (green/yellow) emission filter. Box must be light-tight.
- BC547 pinout is C-B-E (flat face toward you). 2N2222 differs.

LED switched on/off by GPIO25 for background subtraction
Photodiode -> TIA -> ADS1115 -> ESP32 -> TFT
Simplified single-sheet version

Sheet: /
File: Circuit.kicad_sch

Title: ESP32 Handheld Fluorimeter

Size: A4	Date: 2026-10-23	Rev: B
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1